

parsnip: Streamlined Crystallographic Data Parsing for Assembly Science and Engineering

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Summary

parsnip provides a lightweight, precise, and domain agnostic interface for parsing data encoded in the Crystallographic Information File (CIF) 1.1 (Brown & McMahon, 2002), 2.0 (Bernstein et al., 2016), and Macromolecular CIF (mmCIF) (Bourne et al., 1997) formats. Designed for programmatic analysis of crystalline systems, parsnip offers a scriptable API and a suite of convenient data retrieval methods for automated lookups of structural information. Its minimal dependency set and high-level interface allows for the rapid incorporation of parsnip into existing libraries within the molecular simulation ecosystem (Ramasubramani et al., 2020).

parsnip's primary functionality lies in its ability to accurately reconstruct unit cells from simulation data and from experimental data recorded with limited precision, often with only a few decimal places. Through a combination of rational and floating-point arithmetic, we achieve class-leading accuracy when reconstructing large and complex structures. parsnip's detailed processing of structural data yields better alignment with reported space group and point group symmetries than existing tools (Hester, 2006; Larsen et al., 2017; Wojdyr, 2022), providing an ideal foundation for studies centered on material design.

parsnip supports a dictionary-like lookup format for both scalar and tabular data, both of which can be expanded with Unix-style wildcards to simplify complex queries. Convenient methods for parsing unit cell parameters, reconstructing particle positions, and identifying site symmetry data are exposed to streamline common workflows in materials data science. parsnip's clear documentation of conventions and units eliminates ambiguities common to interdisciplinary research. parsnip's use of NumPy structured arrays for data storage simplifies integration into Python, C, and FORTRAN libraries, resulting in a stable, scalable dependency for scientific codebases in materials research at the atomic, molecular, and colloidal scales.

Statement of Need

More than thirty-five years of material data is encoded in the CIF file format, with terabytes of elemental and protein structures freely available to researchers (Gražulis et al., 2009; Ong et al., 2015; wwPDB consortium et al., 2019). While early CIF parsers were predominantly written in C and Fortran, the growth of Python opened new opportunities for simple, scriptable access to crystallographic data. PyCifRW is one of the earliest such tools, offering a complete and specification compliant parser for the CIF format (Hester, 2006). Soon thereafter, ASE, a tool designed to initialize atomistic simulations, added support for CIF files as an alternative to NetCDF inputs for DFT simulations (Bahn & Jacobsen, 2002; Larsen et al., 2017). This marked a transition from pure IO libraries to mixed IO and analysis tools, with both ASE and later projects like pymatgen including code to characterize and manipulate structures (Ong et al., 2013). One of the most recent CIF libraries, gemmi, also follows this trend, although significant amounts of code are committed to quickly and accurately parsing the CIF grammar.

42 While all of these tools provide excellent interfaces for researchers working with atomic materials,
43 the structure and typing of alternative libraries is ill-suited to interdisciplinary research, where
44 the building blocks of crystal structures include atoms, molecules, macromolecules, and
45 nanoparticles. The explosion of simulation research in superatomic systems has driven a need
46 for scalable, array-formatted crystallographic data that easily translates across simulation
47 frameworks and system length-scales. `parsnip` addresses this need by providing a simple,
48 intuitive, and well-documented software frontend that integrates tightly with existing standards
49 for molecular simulation and analysis (Anderson et al., 2020; Thompson et al., 2022).

50 State of the Field

51 `parsnip` supports colloidal and mesoscale materials research in addition to the atomic and
52 protein datasets for which the CIF and mmCIF specifications were originally designed. This is
53 a fundamental change to the scope of the library, and necessitates novel tooling to support the
54 greater diversity of systems. For example, while both `parsnip` and the `gemmi` (Wojdyr, 2022)
55 library support macromolecular data encoded in the mmCIF format, `parsnip` standardizes its
56 API to ensure all inputs are handled in a consistent, programmatic way. In contrast, `gemmi`
57 has separate parsing methods and data structures for each file type, improving performance at
58 the expense of generality.

59 Existing crystallography libraries like `ASE` (Larsen et al., 2017) and `pymatgen` (Ong et al., 2013)
60 encode atomic information into the data structures and types of parsed information, requiring
61 postprocessing for studies where that data is unnecessary or incorrect. Rather than associating
62 pure crystallographic data with atomic symbols or valence states by default, `parsnip` provides
63 only the information required to reconstruct a particular structure unless otherwise queried.
64 This approach offers benefits for users who do require atomic information, as we are able to
65 provide access to arbitrary data associated with the basis positions, rather than fixed keys as
66 required by other tools.

67 Software Design

68 `parsnip`'s design enables studies of crystallinity in colloidal matter, a feature set that is not met
69 by any available tools in the field. By separating units and particle data from pure structural
70 information, we are able to provide a general interface for the study of ordered matter across
71 scales. We also abstract away some portions of the CIF syntax to simplify the API – most
72 notable, we aggregate across data blocks to ensure all relevant information is accessible through
73 a uniform grammar of queries. This choice enables a user interface that more closely resembles
74 other structural data formats like XYZ, MOL, and VTP (MOL File Format, 2025; Schroeder
75 et al., 2006).

76 Our unique approach to the CIF specification extends to the design of our parser as well. While
77 most existing tools in the space use parser generators based on the IUCR's formal grammar, we
78 identified a non-negligible subset of CIF files that break the formal specification but nevertheless
79 contain useful data. To overcome this, `parsnip` does not validate the entire syntax tree of
80 the CIF grammar: rather, we eagerly consume nodes near the leaves of the tree that appear
81 to contain data. Most commonly, this allows `parsnip` to parse data entries with missing or
82 incorrectly escaped delimiters, correctly extracting data that would otherwise be lost. This is a
83 departure from the standard “validating” parser strategy, but it enables fast and robust data
84 extraction without significant increases to code complexity. As detailed in our performance
85 guide, `parsnip` is a complete a standardized CIF benchmark 3 to 100 times faster than most
86 open-source libraries, though significantly slower than the novel `gemmi` parser (Wojdyr, 2022).

87 While our parsing technique alone provides significant accuracy benefits, there are still many
88 files that cannot be accurately reconstructed by other tools. Standard, “symmetrized” CIF
89 data requires the application of symmetry operations to reconstruct a lattice. These operations

are the sum of an experimentally-determined value Wyckoff position and a rational translation, with values wrapped into the range $[0, 1)$. Although Wyckoff positions can have arbitrary real values, the data stored in CIF files is necessarily finite. For this reason, the actual set of valid, parsable positions is the group of rational numbers modulo one. Rather than evaluating expressions in floating point arithmetic like other CIF libraries, parsnip evaluates unit cell positions in the correct rational form. We then convert back to floating point values for a tolerance-based check to remove duplicate atoms, which catches edge cases in recorded data where values are not rounded consistently (e.g. $(1/3, 2/3) \rightarrow (0.3333, 0.6666)$).

Research Impact Statement

parsnip is designed and optimized for integration with larger materials science codes, with a minimal dependency set and pure Python interface. This has facilitated its incorporation into the **freud** analysis library, which uses parsnip to build reference structures for high-throughput simulation analysis (Ramasubramani et al., 2020). When studying complex crystals, standard characterization techniques often fail to uniquely resolve a structure of interest. Simplifying access to a massive array of crystal structures stored in databases like the Crystallography Open Database (COD) (Gražulis et al., 2009) enables the construction of reference datasets for both classical and machine-learned characterization techniques.

We performed tests of parsnip and three other CIF parsing libraries against a subset of the Crystallography Open Database (COD), choosing the correct structure to be the most frequently reported value across the four packages. Table 1 shows that parsnip's rational parsing approach was able to correctly extract 96.5% of structures, more than any other library we could find. We note that our results use a single, fixed parsing precision for all 10,094 files. However, as discussed in parsnip's documentation, tailoring the parse precision to match the precision of the data in the file yields even better results.

Table 1: Comparison of unit-cell reconstruction consensus for 10094 CIF files from the COD. "Total Correct" indicates the total number of correctly-reconstructed crystals and "Failed to Parse" indicates files that could not be read at all.

Library	Correct Crystals	Incorrect Crystals	Failed to Parse	Percent Correct
parsnip	9740	17	337	96.5%
ASE	9247	37	810	91.6%
pymatgen	9255	35	804	91.7%
gemmi	8277	1817	0	82.0%

parsnip equals or outperforms the tested suite of libraries in accurately reproducing the space group of crystal data. Table 2 demonstrates this efficacy, using the 9112 CIF files from the previous test that included explicit space group information. Using `spglib==2.7.0` with `symprec=1e-3`, the reconstructions are evaluated against the original file's keys and classified into four categories: "Correct Space Group" (matches the original record); "Symmetry Too High" (often due to incorrect rounding); "Symmetry Too Low" (often due to extraneous sites); and "Undetermined" (no space group identified at the given precision). We note that while parsnip does not have the lowest total number of "Symmetry Too High" cases, the additional errors primarily occur where numeric placeholder values like `-1` are present, as we treat such entries as valid data where other tools do not.

Table 2: Comparison of space-group accuracy for 9114 CIF files from the COD with stored space-group data.

Library	Correct Space Group	Symmetry Too High	Symmetry Too Low	Undetermined
parsnip	8800	105	43	164
ASE	8471	106	357	178
pymatgen	8108	98	420	486
gemmi	8359	82	46	625

parsnip also supports Unix-style wildcard queries, simplifying common lookup patterns like cell parameter extraction and space group identification. Single-character wildcards enable specification-compliant queries into heterogeneous databases of both CIF and mmCIF files, further accelerating programmatic materials exploration. Most importantly, this allows users to extract equivalent data entries whose naming depends on specification. For example, there are many cases where the mmCIF key-naming convention differs from standard CIF files by a single character. The following code block shows two examples where this wildcard syntax simplifies user scripts. Although the `gemmi` library does support a similar style of wildcard through their `gemmi grep` command-line tool, its use is limited to bash scripting, and each wildcard query requires the file to be re-parsed in its entirety (Wojdyr, 2022).

```
from parsnip import CifFile

crystal = CifFile("my_data.cif")
protein = CifFile("protein.mmCIF")

cif_cell_keys = ("_cell_length_a", "_cell_length_b", "_cell_length_c")
mmc_cell_keys = ("_cell.length_a", "_cell.length_b", "_cell.length_c")

# Wildcard to extract all cell lengths
assert all(crystal[cif_cell_keys] == crystal["_cell_length*"])

# Wildcard to extract the same data from CIF and mmCIF
assert all(protein[mmc_cell_keys] == protein["_cell?length*"])
assert all(crystal[mmc_cell_keys] == crystal["_cell?length*"])
```

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AI Usage Disclosure

GitHub Copilot's automated PR review was evaluated for use with this project, but was disabled due to the unhelpful nature of the output. No generative AI tools were used in the development of this software, the writing of this manuscript, or the preparation of supporting materials.

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